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Standard Library Hardening - using `std::optional<T&>`

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Abstract

Utilize `std::optional<T&>` ^[1] in `inplace_vector` ^[2] in order to harden it against null pointer dereference errors for similar memory safety reasons as Standard Library Hardening ^[3] and Minor additions to C++26 standard library hardening ^[4].

This was discussed during the standardization of `inplace_vector` ^[2:1] but could not be adopted since `std::optional<T&>` ^[1:1] was not sufficiently along in the C++26 standardization process. Now that `std::optional<T&>` ^[1:2] is being considered during the June 2025 Sophia meeting, it would be ideal that `inplace_vector` ^[2:2] underwent minor revisions to better align it with the rest of C++26 and consequently provide a more optimum interface.

What is being asked for is that 3 of `inplace_vector`'s ^[2:3] modifiers be revised to return an `std::optional<T&>` ^[1:3] instead of a pointer.

```
template< class... Args >
constexpr pointeroptional<T&> try_emplace_back( Args&&... args );

constexpr pointeroptional<T&> try_push_back( const T& value );

constexpr pointeroptional<T&> try_push_back( T&& value );
```

Motivation

Returning a pointer from these three methods are less than an ideal interface. The primary problem is that it leads to null pointer dereference errors. It is essentially a 100% unsafe interface. Now contrast that with the different iterations of `optional`.

While `optional<T>` can't be used as an alternative to pointer, in C++17 when it was standardized, `optional` had an equal number of unsafe and safe modifiers. The unsafe ones were `operator->` and `operator*` and the safe modifiers were `value` and `value_or`. In other words, `optional` was born with 50% null pointer dereference safety.

In C++23, the `and_then`, `transform` and `or_else` modifiers were added. This brought the total of modifiers to 5 of 7. The result is that 71% of all modifiers are safe from null pointer dereference errors.

However, only in C++26 with the advent of `std::optional<T&>` ^[1:4] does `optional` becomes a viable substitute for a nullable pointer. Also in this release is `std::optional` range support ^[5] which turns `optional` is a range of 0 or 1. This opens the door for 125 range algorithms and 34 range adapters to become null pointer deference safe modifiers for `optional`. Those range algorithms and adapters are growing at a much faster rate than new modifiers being added to the `optional` class itself. This brings the modifier safety rating up to 98%. The original 2 unsafe modifiers and hence the remaining 2% are being addressed by Standard Library Hardening ^[3:1].

Besides the safety benefit, the monadic and range usage are more ergonomic than the legacy pointer based interface.

23.3.16 Class template `inplace_vector` [`inplace.vector`]

23.3.16.1 Overview [`inplace.vector.overview`]

...

(5.3.3) — If

...

```
namespace std {  
    template<class T, size_t N>  
    class inplace_vector {  
    public:
```

...

```
template< class... Args >  
constexpr pointeroptional<T&> try_emplace_back( Args&&... args );  
  
constexpr pointeroptional<T&> try_push_back( const T& value );  
  
constexpr pointeroptional<T&> try_push_back( T&& value );
```

...

```
};  
}
```

...

23.3.16.5 Modifiers [`inplace.vector.modifiers`]

...

```
template<class... Args>  
constexpr pointeroptional<T&> try_emplace_back(Args&&... args);  
constexpr pointeroptional<T&> try_push_back(const T& x);  
constexpr pointeroptional<T&> try_push_back(T&& x);
```

⁸ Let `vals` denote a pack:

(8.1) — `std::forward<Args>(args)...` for the first overload,

(8.2) — `x` for the second overload,

(8.3) — `std::move(x)` for the third overload.

⁹ *Preconditions*: `value_type` is Cpp17EmplaceConstructible into `inplace_vector` from `vals...`.

¹⁰ *Effects*: If `size() < capacity()` is true, appends an object of type `T` direct-non-list-initialized with `vals...`. Otherwise, there are no effects.

¹¹ *Returns*: `nullptr` if `size() == capacity()` is true, otherwise `addressof(back())`.

¹² *Throws*: Nothing unless an exception is thrown by the initialization of the inserted element.

¹³ *Complexity*: Constant.

¹⁴ *Remarks*: If an exception is thrown, there are no effects on `*this`.

Impact on the standard

Other than `inplace_vector` ^[2:4], there are no other changes to the library standard and since C++26 has not been released, this tweak is not expected to cause any problems.

The proposed changes are relative to the current working draft N5008 ^[6].

References

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